# Wasi M.F. Naqvi

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## Education

2026– **PhD Physics, McGill University** GPA: 4.00/4.00
Supervisor: Nicolas B. Cowan
Thesis: A Patchwork of Worlds: Artificial Intelligence for Population-Level Studies of Exoplanet Atmospheres

2025–26 **M.Sc. Physics, McGill University** GPA: 4.00/4.00
Supervisor: Nicolas B. Cowan
Thesis: HERMES: HiERarchical Modelling for Exoplanet Science

2021–25 **B.Sc. Physics (Honours), The University of British Columbia** GPA: 4.00/4.33
Minor in Data Science and Statistics
Thesis: Simulating In-Orbit Performance for the CASTOR Mission
Supervisor: Tyrone Woods
Distinctions: Dean's List, Best Thesis Award

## Publications

**Naqvi, W. M. F.** and Cowan, N. B. ***HERMES: HiERarchical Modelling for Exoplanet Science.*** Royal Astronomical Society, Techniques and Instruments (2026,Submitted, under review.). Available on arxiv

Côté et al. (including **Naqvi, W.**). ***The CASTOR Mission***. Special Issue, Journal of Astronomical Telescopes, Instruments, and Systems, SPIE. (2025)

## Awards & Honours

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|---|---|---|
| 2025- | Kharusi Family International Science Fellowship, McGill University | $15,000 CAD |
| 2021–25 | Karen McKellin International Leader of Tomorrow Award, UBC | $250,000 CAD |
| 2025 | McGill Graduate Excellence Award | $5000 CAD |

## Contributed Talks & Posters

♁ Contributed Talk ♃ Poster

♁ Too Much Universe, Not Enough Grad Students: Deep Learning in Astrophysics — Aug 2026
Trottier Space Institute, McGill University — Montréal, Canada

♃ HERMES: HiERarchical Modelling for Exoplanet Science — Jun 2026
CASCA Annual General Meeting. — Montréal, Canada

♁ Comparative climatology of exoplanets using Hierarchical Modelling, — May 2026
Trottier Institute for Research on Exoplanets — Montréal, Canada

♁ **HERMES: HiERarchical Modelling for Exoplanet Science** — Mar 2026
Ariel Open Conference '26 — ESA Harwell, UK

♁ BayesBall: Bayesian Modelling for sport analytics, Nov 2025
Trottier Space Institute Lunch Talks Montréal, Canada

♁ A detector simulation pipeline for the CASTOR mission **(Invited)** Jun 2026
The CASTOR Mission Consortium, NRC Herzberg Astronomy & Astrophysics Victoria, Canada

## Research & Work Experience

**Trottier Institute for Research on Exoplanets, McGill University** Montréal, QC
*Summer Researcher* May 2025–Present
Hierarchical Bayesian Modelling of planetary targets to probe population-level trends for the Ariel Space Mission.

**NRC Herzberg Astronomy & Astrophysics** Victoria, BC
*Junior Adaptive Optics Research Scientist* Jan 2025–Present
Implementing a convolutional neural network on an Adaptive Optics test bench for the 1.2 m REVOLT Telescope at the Dominion Astrophysical Observatory. Running machine-learning simulations alongside lab experiments under Dr. Maaike van Kooten.

**TRIUMF** Vancouver, BC
*Universal Polarization Research Assistant* Sep 2024–Present
Devising a Faraday Rotation method to measure polarization of optically-pumped Rb vapour at ISAC-II. Writing numerical simulations for polarizer experiments and developing a universal laser-polarization method for isotopes with unknown atomic transitions.

**NRC Herzberg Astronomy & Astrophysics** Victoria, BC
*Junior Research Scientist* May–Aug 2024
Designed Python software pipelines for imaging-sensor simulations and detector modelling for the CASTOR mission (Canadian Space Agency / NRC). Built with Dask (parallel computing) and ESA Pyxel (detector physics).

**COCANA Lab, UBC Okanagan** Kelowna, BC
*Undergraduate Research Assistant* May–Aug 2023
Optimization research for a commercial road-building project (SoftTree Inc.) under Dr. Yves Lucet. Used YALMIP solver; translated codebases between MATLAB and Python with unit-test verification.

**CAN-SBX Stratospheric Balloon Design Team** Kelowna, BC
*Physics & Data Science Lead* Sep 2023–May 2024
Led physics instrumentation and data analysis. Developed efficient MicroPython / C++ firmware for flight micro-controllers, designed ML models for atmospheric prediction, and assisted with dual-polarising antenna calibration and circuit analysis.

## Teaching Assistantships

**PHYS 180** Space, Time and Matter Sep-Dec 2025
**PHYS 241** Signal Processing Jan-Apr 2025

## Technical Competitions

Primary role across competitions: building, packaging, and benchmarking Bayesian models, U-Nets, Transformers, and CNNs for scientific data challenges.

ExoHack: Ariel Machine Learning Hackathon (**Winner)** Mar 2026
NeurIPS Ariel Data Challenge Jun 2026
Rocky Worlds DDT Challenge May 2026

## Community Engagement & Outreach

Founding Member & Organizing Committee, Montréal AstroStats Day 2026–

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|---|---|
| Author, Astrobites | 2025– |
| Mentor & Teaching Assistant, Small Private Online Courses, Scientists for Displaced People | 2026– |
| iREx EDI Committee Member | 2025– |
| McGill Hackathon Mentor | 2026– |
| Astro on Tap Volunteer | 2026– |
| Volunteer, McGill Physics "World of STEM" high-school astronomy sessions | 2025– |
| Coordinator, graduate student lunches with TSI seminar speaker | 2025– |
| Facilitator, McGill "From Planets to Particles" family science outreach fair | 2025– |
| Organizing & Administration Volunteer, Exoclimes 2025 Conference | 2025- |
| Science Outreach, Trottier Institute for Research on Exoplanets | 2025– |
| UBCO Physics Student Club Treasurer | 2021–2025 |
| Exercise is Medicine Coordinator | 2023–2025 |
| Volunteer Soccer Coach, Royal Oak, Victoria | 2024–2025 |
| Mentor, CERN BeamLine For Schools (BL4S), Cedar College, Karachi | 2021– |
| Science Outreach, Let's Talk Science Okanagan | 2023–2024 |
| Volunteer, Okanagan Food Bank | 2022–2024 |
| Volunteer, Student Union Okanagan, UBC | 2022–2023 |

## Skills

**Programming:** Python, C, C++, Julia, Fortran, JavaScript, R, SQL, Shell/Bash
**Markup & Typesetting:** LaTeX, HTML/CSS, Markdown
**Python Ecosystem:** PyTorch, JAX, NumPyro, PyMC, Emcee, NumPy, SciPy, Astropy, Scikit-learn, Dask, Matplotlib, Pandas, ArviZ, Corner, Dynesty
**Statistical & ML Methods:** Hierarchical Bayesian Modelling, Gaussian Processes, MCMC Sampling, Nested Sampling, Simulation-Based Inference, PCA, CNNs, U-Nets, Transformers, Variational Inference, Model Comparison and Bench Marking
**N-Body Codes:** REBOUND
**Computing Infrastructure:** CANFAR, DRAC Supercomputing Clusters, Slurm, MPI, Git/GitHub, Docker, Linux
**Other Software:** Microsoft Excel, Google Sheets

## Languages

**Fluent:** English, Urdu, Farsi (Persian)
**Conversational:** Hindi, Dari, Punjabi
**Basic:** French, Arabic

## Relevant Coursework

Radiative Processes in Astrophysics, Statistical Learning, Computational Astrophysics, Machine Learning Theory, Probability Theory, Stochastic Processes, Exoplanets and Brown Dwarfs.